

KIRUBIA A

Data Analyst · Machine Learning · Power BI & Tableau

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PROFESSIONAL SUMMARY

Results-driven Data Analyst & ML practitioner with a B.Tech in AI & Data Science (CGPA 8.2) and hands-on experience building end-to-end ML pipelines, ETL workflows, interactive dashboards, and business-impact analyses on datasets exceeding 150,000 records. Delivered a 12% accuracy improvement in production models, reduced stakeholder ad-hoc reporting time by ~30%, and translated complex findings into executive-ready insights. Proficient in Python, SQL, Power BI, and Tableau with supporting ML expertise (XGBoost, SHAP, MLflow) and GCP (BigQuery) exposure for cloud-based data processing.

TECHNICAL SKILLS

Languages	Python (Pandas, NumPy), SQL (MySQL — joins, window functions, CTEs, subqueries), Excel (Pivot Tables, XLOOKUP)
Visualisation	Power BI (DAX, KPI cards, slicers, drill-through, calculated columns), Tableau (LOD expressions, calculated fields), Matplotlib, Seaborn, Plotly, Streamlit
ML & AI	Scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, K-Means, SHAP (model explainability), SMOTE
NLP	NLTK, spaCy, TF-IDF vectorisation, VADER Sentiment Analysis
Cloud & Tools	GCP (BigQuery, Colab), MLflow (experiment tracking), Git & GitHub, Jupyter Notebook, VS Code
Core Concepts	ETL Pipelines, Data Modelling, EDA, Feature Engineering, Statistical Analysis, Hypothesis Testing, A/B Analysis, Data Storytelling, Business Intelligence

PROFESSIONAL EXPERIENCE

Data Science Intern | Gradtwin Services | Sept 2024 – Nov 2024

Contributed to a client-facing data science engagement, owning the full data pipeline from raw ingestion through to stakeholder reporting.

- Engineered end-to-end ETL pipeline with feature engineering (label encoding, one-hot encoding, normalisation) and hyperparameter tuning via GridSearchCV — boosting production model accuracy by 12%.
- Built and benchmarked ML models (KNN, Random Forest) using 5-fold cross-validation; selected best-fit model using F1-score, ROC-AUC, precision, and recall — reducing misclassification rate.
- Conducted EDA on real-world multi-feature datasets using Pandas, Matplotlib, and Seaborn; surfaced key outliers, distribution skews, and feature correlations that directly shaped client-facing recommendations.
- Delivered 4+ stakeholder-ready BI reports translating technical model outputs into clear narratives with annotated charts, KPI summaries, and actionable takeaways for non-technical decision-makers.
- Collaborated with senior data scientists on feature selection strategies, contributing to reduced model complexity without sacrificing predictive performance.

KEY PROJECTS

Nykaa Beauty Brand Review Analysis — Python · Pandas · SQL · Matplotlib · Seaborn · VADER · Power BI

End-to-end analytics project transforming raw customer review data into brand intelligence for a major e-commerce beauty platform.

- Engineered ETL pipeline ingesting and cleaning 61,284 customer reviews across 11 brands — handling nulls, duplicates, encoding issues, and schema standardisation before analysis.
- Wrote complex SQL queries (multi-table joins, GROUP BY aggregations, window functions) to segment reviews by brand, rating tier, and time period.
- Applied VADER sentiment scoring across all 61,284 reviews; cross-validated sentiment polarity against star ratings to identify brand-specific sentiment drift — surfaced Kay Beauty as category leader with a 4.69 avg rating and highest 5-star ratio, directly informing brand partnership strategy.
- Designed a 9-visual interactive Power BI dashboard featuring brand slicers, dynamic KPI cards (avg rating, review volume, sentiment score), and trend charts — reduced stakeholder ad-hoc reporting requests by ~30%.
- Delivered written insight report to stakeholders highlighting under-performing brands, seasonal review spikes, and product categories driving negative sentiment.

Global AI Adoption & Workforce Impact — Python · XGBoost · LightGBM · K-Means · SHAP · MLflow · Streamlit · SMOTE

Large-scale ML and analytics project examining how AI adoption is reshaping workforce structures across 20+ global industries.

- Designed and executed a full ML pipeline on 150,000+ records spanning 20+ industries — data cleaning, feature engineering, SMOTE-based class balancing, XGBoost multi-class classification, and K-Means workforce segmentation.
- Integrated SHAP explainability to surface the top drivers of AI adoption by industry, making complex model outputs interpretable for non-technical business stakeholders.
- Tracked 12+ experiment runs using MLflow (parameters, metrics, and model artefacts), enabling systematic comparison and accelerating the modelling iteration cycle.
- Deployed a real-time interactive Streamlit dashboard with industry filters, adoption score displays, and cluster visualisations — enabling independent exploration by analysts and decision-makers without requiring code access.

Sentiment Analysis — Product Reviews — Python · NLTK · spaCy · TF-IDF · Logistic Regression · Scikit-learn

End-to-end NLP classification system built to categorise product review sentiment at scale.

- Built full NLP pipeline: text cleaning → tokenisation → stopwords removal → lemmatisation → TF-IDF vectorisation → Logistic Regression classifier.
- Evaluated model rigorously using precision, recall, F1-score, and confusion matrix — achieving production-grade classification performance across sentiment classes.
- Compared TF-IDF + Logistic Regression against baseline approaches; documented performance trade-offs across vectorisation strategies.

Sales Data Analysis & KPI Dashboard — Python · SQL · Tableau · Excel · Pandas · Seaborn

Business analytics project focused on identifying revenue leakage and building a live KPI reporting solution for a sales team.

- Analysed 9,800+ sales transaction records using SQL and Python; performed cohort analysis, discount-tier segmentation, and deal velocity calculations to map revenue trends across product lines and regions.
- Identified excessive and inconsistent discounting as a significant net revenue drain — quantified the impact and packaged findings into a corrective pricing recommendation deck presented to stakeholders.
- Built a multi-view Tableau KPI dashboard visualising revenue trends, average deal size, discount rate vs. close rate correlation, and pipeline velocity — adopted as the sales team's primary reporting tool, replacing manual Excel reporting.
- Automated data refresh pipeline using Python (Pandas) to clean and reshape weekly sales exports, cutting dashboard prep time significantly.

Cirrhosis Disease Stage Prediction — Python · Scikit-learn · Random Forest · Logistic Regression · SMOTE

Clinical ML project predicting disease progression stages from patient diagnostic data.

- Trained multi-class classifier on clinical data; Random Forest outperformed baseline by 18% — selected as production model after rigorous cross-validated evaluation.
- Applied SMOTE to resolve class imbalance across disease stages; compared Random Forest vs. Logistic Regression using stratified accuracy, F1-score, and AUC-ROC metrics.
- Analysed feature importance to identify the strongest clinical predictors of disease progression, providing interpretable outputs for potential medical use.

EDUCATION

B.Tech — Artificial Intelligence & Data Science | Saveetha School of Engineering, Chennai | 2021 – 2025

CGPA: 8.2 / 10 | Relevant Coursework: Machine Learning, Deep Learning, NLP, Statistics, Database Systems, Data Visualisation

CERTIFICATIONS

- Accenture Data Analytics & Visualisation Job Simulation — Forage | Aug 2024
- Oracle Cloud Infrastructure GenAI Certified Professional — Oracle | Jul 2024
- Oracle Cloud Infrastructure AI Foundations Associate — Oracle | Aug 2024